Physics-Residual Test-Time Adaptation for Graph Neural Network PDE Surrogates

Label-free correction of MeshGraphNet predictions using the governing equation

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Neural surrogates are fast — but brittle

- Numerical PDE solvers are accurate but expensive per query.
- A **surrogate** network learns the solver's one-step map $u_t \mapsto u_{t+1}$ and answers in milliseconds.
- Under **distribution shift** — new viscosity, distorted mesh, unseen initial wave — accuracy drops.
- Retraining needs new labelled simulations, which are rarely available at deployment.

Research question

Can a trained surrogate correct itself *at test time*, using only the input and the known physics, with **no ground-truth labels**?

Idea: a correct prediction satisfies the PDE. Its *residual* is a free, label-free error signal.

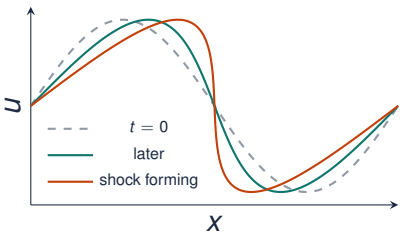
Contributions

1. A MeshGraphNet surrogate for the 1D viscous Burgers' equation (PDEBench).
2. A differentiable finite-difference residual for uniform and non-uniform meshes.
3. A decoder-only test-time adaptation procedure with anchor regularisation and best-state selection — no labels used.
4. A **residual gate** that runs TTA only when the physics check can still see an error, calibrated on separate data.
5. A held-out evaluation: **20 unseen trajectories**, mean $\pm$ std, paired Wilcoxon tests; plus ablations and geometric shift.
6. A 2D extension on **trained** FNO-2D / U-Net-2D surrogates (8 datasets) and an interactive dashboard (Next.js + Express + PyTorch).

Governing equation: 1D viscous Burgers'

$$\underbrace{\frac{\partial u}{\partial t}}_{\text{time change}} + \underbrace{u\frac{\partial u}{\partial x}}_{\text{advection}} = \underbrace{\nu\frac{\partial^2 u}{\partial x^2}}_{\text{diffusion}} \qquad x \in [0,1] \text{ (periodic)}, \quad \nu = 10^{-3}$$

- **Advection** steepens the wave: fast regions overtake slow ones.
- **Diffusion** smooths it; small $\nu \Rightarrow$ near-discontinuous shocks.
- Minimal model with the two hard ingredients of fluid flow.



Dataset and graph construction

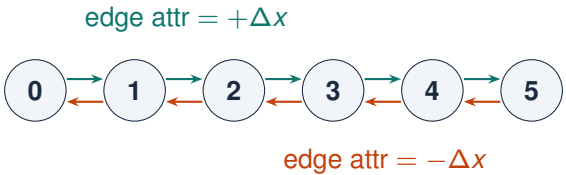
PDEBench 1D Burgers, $\nu = 0.001$

`1D_Burgers_Sols_Nu0.001.hdf5`

- `tensor`: [trajectories, time, 1024 grid points]
- $\Delta t = 0.01$, $\Delta x \approx 1/1024$
- One-step pairs: input u_t, target u_{t+1}

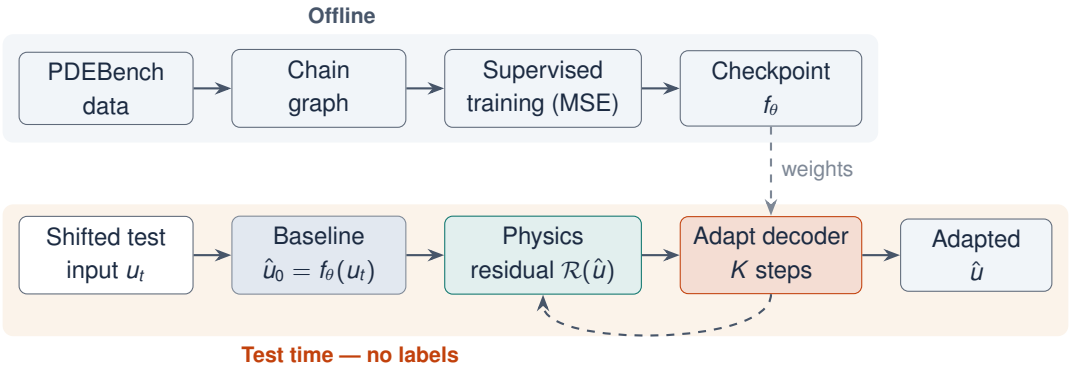
Graph (PyTorch Geometric `Data`)

- 1024 nodes, 2046 directed edges
- node feature $= u_t$, edge feature $= \pm\Delta x$

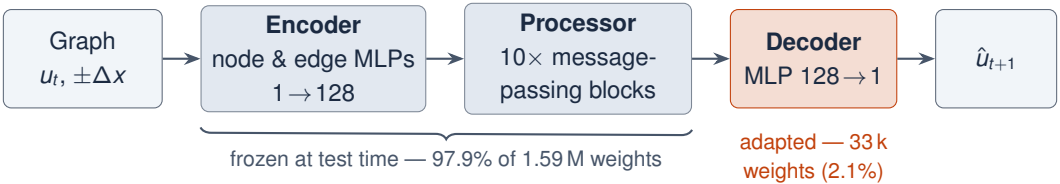


each grid point is a node;
neighbours exchange messages along edges

Pipeline overview



Surrogate architecture: MeshGraphNet (Pfaff et al., ICLR 2021)



$$\mathbf{e}_{ij} \leftarrow \mathbf{e}_{ij} + \mathrm{MLP}_e(\mathbf{v}_i, \mathbf{v}_j, \mathbf{e}_{ij}) \qquad \mathbf{v}_j \leftarrow \mathbf{v}_j + \mathrm{MLP}_v\left(\mathbf{v}_j, \textstyle\sum_{i \in \mathcal{N}(j)} \mathbf{e}_{ij}\right)$$

Residual updates; latent width 128; ReLU MLPs with LayerNorm; receptive field of 10 hops.

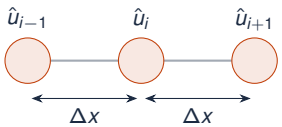
Label-free signal: the discrete physics residual

Finite differences on the prediction $\hat{u}$ (periodic neighbours):

$$\partial_t u \approx \frac{\hat{u}_i - u_{t,i}}{\Delta t}$$

$$\partial_x \hat{u} \approx \frac{\hat{u}_{i+1} - \hat{u}_{i-1}}{2\Delta x}$$

$$\partial_{xx} \hat{u} \approx \frac{\hat{u}_{i+1} - 2\hat{u}_i + \hat{u}_{i-1}}{\Delta x^2}$$



Non-uniform meshes use local gaps $h_\pm$:

$$\partial_x u \approx \frac{u_{i+1} - u_{i-1}}{h_+ + h_-}$$

$$\mathcal{R}_i = \partial_t u + \hat{u}_i\, \partial_x \hat{u} - \nu\, \partial_{xx} \hat{u}, \qquad \mathcal{L}_{\text{phys}} = \frac{1}{N} \sum_i \mathcal{R}_i^2 \quad \text{(differentiable in PyTorch)}$$

Test-time adaptation algorithm

Input: test graph $(u_t, \mathcal{G})$, trained f_θ, K, η, λ

$\hat{u}_0 \leftarrow f_\theta(u_t)$; `// baseline / anchor`

freeze encoder + processor; $\mathbf{h} \leftarrow$ latent (cached once);

save decoder weights $\theta_d^{(0)}$; $\theta_d^\star \leftarrow \theta_d^{(0)}$;

for $k = 1$ **to** K **do**

  $\hat{u} \leftarrow \mathrm{Dec}_{\theta_d}(\mathbf{h})$;

  $\mathcal{L} \leftarrow \mathcal{L}_{\text{phys}}(\hat{u}, u_t) + \lambda \, \|\hat{u} - \hat{u}_0\|^2$;

  clip $\|\nabla\| \leq 0.5$; Adam step on θ_d;

  if $\mathcal{L}_{phys}$ *is lowest so far* **then** $\theta_d^\star \leftarrow \theta_d$;

return $\mathrm{Dec}_{\theta_d^\star}(\mathbf{h})$; restore $\theta_d^{(0)}$;

Hyper-parameter	Value
Steps K	12
Learning rate η	8×10^{-5}
Anchor λ	0.1
Grad. clip	0.5
Optimiser	Adam

Ground truth is used **only** afterwards, to score the result.

Residual gate: adapt only when the physics sees an error

A low residual does not always mean a low error. If the model is already better than the residual can resolve, TTA only adds noise.

- Label-free score:
  $$q = \frac{R(\hat{u})}{R(\text{copy } u_t)}$$
- Floor q_{floor}: median q of the *true* next state on calibration trajectories (per mesh).
- Adapt iff $q > \kappa \cdot q_{\text{floor}}$.
- $\kappa \in \{1, \ldots, 5, \infty\}$ chosen on calibration data; ∞ = never adapt.

Split (1D). Train: trajectories 0–8999. Calibrate: 9900–9919. Test: 9980–9999. No test labels are used to pick κ.

Also fixed: the best-state snapshot was saved *after* the optimiser step it was scored on. It is now saved before, and the final state is checked too.

Crank–Nicolson residual: truth scores $0.18\times$ the copy residual ($0.77\times$ for implicit), so it is the sharper check.

Held-out evaluation: 20 unseen trajectories

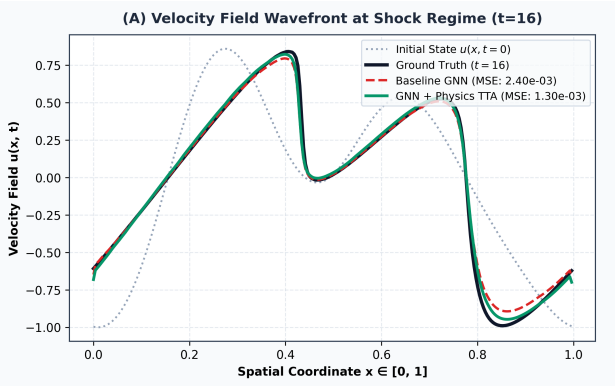
100 samples per mesh (20 trajectories $\times$ 5 timesteps). Original model, 12 steps, $\eta = 10^{-4}$, $\lambda = 0.1$.

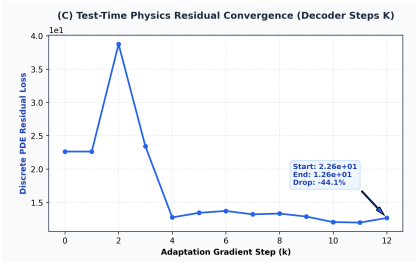
Mesh	Baseline rel. L_2	Plain TTA	**Gated TTA**	Worse (plain→gated)	p (gated)
Uniform	9.62%	+15.9% $\pm$ 19.4	**+18.9%** $\pm$ 15.4	11% $\rightarrow$ 4%	1.3×10^{-5}
Jitter $\pm$35%	9.60%	+14.4% $\pm$ 25.4	**+18.1%** $\pm$ 13.5	8% $\rightarrow$ 0%	2×10^{-6}
Warped	9.49%	+14.0% $\pm$ 26.9	**+18.0%** $\pm$ 16.1	8% $\rightarrow$ 0%	2×10^{-6}
Coarse N=512	9.57%	+22.7% $\pm$ 16.1	**+22.7%** $\pm$ 16.1	2% $\rightarrow$ 2%	2×10^{-6}

Every gated gain is significant (paired two-sided Wilcoxon over trajectories, $p \leq 1.3 \times 10^{-5}$), and the gate also **cuts the spread**: std 13–16% vs. 19–27% for plain TTA. Earlier we quoted 73.9% from 3 hand-picked samples; *these* are the numbers to quote.

Error reduction = MSE change per trajectory, mean $\pm$ std over the 20 trajectories (positive = TTA helped).

Qualitative result at a shock

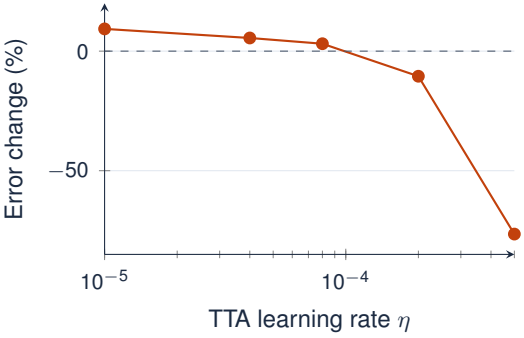




	MSE	Residual
Baseline	2.40×10^{-3}	22.6
After TTA	1.30×10^{-3}	12.6
Change	-45.6%	-44.1%

Correction is concentrated at the steep front ($x \approx 0.8$). The spike at step 2 is why we keep the best state, not the last.

Ablations: learning rate and adaptation scope



Scope (10 steps)	Error	Time
Decoder only	−11.4%	0.34 s
Full network	−136.8%	1.37 s

- Small step sizes are essential; $\eta \geq 2 \times 10^{-4}$ overshoots.
- Adapting the full GNN is **4**× **slower** and far less stable.

4 held-out samples; error change relative to unadapted model (negative = worse).

Geometric shift and an accurate model

Geometric shift is no longer an open problem. The earlier losses (+13.8% MSE on warped) came from the best-state bug. After the fix, plain TTA on the original model:

Shift	Error reduction
Jitter ±35%	+34.9%
Warped	+28.0%
Coarse (N=512)	+36.9%

Same samples as the original table; the held-out table on slide 10 is the reliable version.

v2 model (delta prediction, normalisation; 2.35% one-step error vs. 12.8%):

Mesh	Plain TTA	**Gated**
Uniform	−190.6%	−8.2% (p=0.06)
Jitter	−61.9%	0% (shut)
Warped	+13.7%	**+13.1%**
Coarse	+36.9%	**+36.9%**

Plain TTA hurts 82–89% of v2 samples on uniform and jitter; the gate cuts that to $\leq$5% and keeps the gains where v2 is weak.

2D results: the gate decides where TTA is safe

Trained FNO-2D / U-Net-2D on 8 datasets (5 Lattice-Boltzmann obstacle flows, heat, advection, vorticity). Per dataset: 6 train, 1 calibration, 3 test trajectories (57 test frames); κ and q_{floor} from the calibration trajectory only.

Dataset	Model	Plain TTA	**Gated TTA**	Frames worse (plain→gated)
5 LBM obstacle flows	FNO-2D	$+0.9\%$ to -20.2%	0% (gate shut)	35–100% → **0%**
5 LBM obstacle flows	U-Net-2D	-11.9% to -249%	0% (gate shut)	55–100% → **0%**
Heat	FNO / U-Net	+25.0% / +45.4%	**+25.0%** / **+45.4%**	0% → 0%
Advection	FNO / U-Net	+6.6% / +67.5%	**+6.6%** / **+67.5%**	0% → 0%
Vorticity	FNO / U-Net	+5.7% / +57.7%	**+5.7%** / **+57.7%**	0% → 0%

The gate opened on **every** frame of the three equations with an exact residual ($p \approx 5 \times 10^{-11}$ over frames) and on **no** frame of the LBM flows, where the trained models already beat what the residual can resolve. No 2D test frame got worse.

Frames within a trajectory are correlated, so treat 2D p-values as indicative.

Interactive system



- Run TTA on any trajectory and timestep; view baseline vs. adapted vs. truth side by side.
- Live training telemetry; dataset explorer with space–time plots; 2D flow visualiser.

Limitations & future work

Limitations

- Gains depend on the residual: TTA only helps where the PDE check is sharper than the model's error.
- The gate protects, but it cannot make TTA help an already-accurate model (v2 uniform, LBM flows).
- Needs the governing PDE and its parameters.
- $\sim$0.35 s overhead per sample on CPU.
- 2D tests: 3 trajectories; frames correlated.
- Single-step evaluation; v2 rollouts unstable.

Future work

- Sharper residuals for LBM flows (e.g. lattice-level consistency) so the gate can open there.
- More 2D test trajectories for trajectory-level tests.
- Shift in the data (true $\nu = 0.002$ simulations).
- Trajectory-level TTA for long autoregressive rollouts.
- Adapt edge encoder / LayerNorm under geometric shift.

Conclusion

A trained surrogate can be corrected at test time without labels by minimising the PDE residual through its decoder — and a label-free gate tells us when not to.

- **Held-out 1D:** +18–23% lower error on all four meshes (20 unseen trajectories, $p \leq 1.3 \times 10^{-5}$), with $\leq$4% of samples made worse.
- **Geometric shift:** works after the best-state fix (+28–37%).
- **2D:** +45–68% (U-Net) on heat, advection and vorticity; the gate stays shut on obstacle flows, so no frame got worse.
- Decoder-only adaptation is 4$\times$ cheaper and far more stable than full fine-tuning.
- TTA is only as good as its physics residual — the gate measures that directly.

References

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Thank you

Questions?